

Gerald Emanuel Vargas Morales

Computer Engineering Student · Full Stack Developer · Data Analysis · Applied AI

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📍 Cochabamba, Bolivia



PROFILE

Final-year Computer Engineering student at **Universidad Mayor de San Simón (UMSS)** with solid technical training and **2+ years of experience delivering paid projects** for businesses and institutions. Contributed to the development of **electoral systems** for the Cochabamba Sub-national Elections, mobile applications for the Municipal Government of Colcapirhua, management systems for local businesses, and modern educational platforms. Experienced in **deploying and maintaining multiple production systems on Linux servers**. Committed to technical excellence and continuous learning, with a strong focus on **Artificial Intelligence** and **Machine Learning**.

PROFESSIONAL EXPERIENCE

Full Stack Developer — Official Internship

January 2026 – April 2026

Gobierno Autónomo Municipal de Colcapirhua · Laravel 11 + Flutter 3.41

- ▷ Developed **ColcaTrufis**: cross-platform urban mobility app (Trufi/Radiotaxi) with official municipal route data.
- ▷ Stack: **Flutter** (mobile frontend) + **Laravel** (RESTful APIs). Development and maintenance of institutional websites.
- ▷ Digital solutions for optimising municipal administrative processes.

Full Stack Developer — Electoral Ballot System (NGP Party)

December 2025 – March 2026

Sub-national Elections Cochabamba · Node.js + React

CURRICULAR CERTIFICATE

- ▷ **Successfully deployed** and used during the official Cochabamba sub-national electoral process.
- ▷ Backend: **Node.js/Express** — RESTful APIs for real-time electoral data processing.
- ▷ Frontend: **React** — interactive dashboard for live visualisation of results. Scalable clean architecture.

Full Stack Developer — Inventory & Management System

2026 · In Production Use

Ferretería Urkupiña · Laravel 12 · React · Tailwind CSS

PAID PROJECT

- ▷ Designed and developed a complete **inventory and product management system** for a hardware store, currently **in active use** by the business.
- ▷ Backend: **Laravel 12**. Frontend: **React + Tailwind CSS** — custom dashboard with interactive stats cards, product tracking and stock control.
- ▷ Focused on clean UI/UX with custom animations and a cohesive brand identity for day-to-day operational use.

Full Stack Mobile Developer — Electricity Meter Reading System

2024

Cooperativa de Electricidad “15 de Noviembre” — Potosí, Bolivia · Flutter · Laravel · ESC/POS · Bluetooth · OpenStreetMap

PAID PROJECT

Android mobile application developed for field technicians of the “**15 de Noviembre**” electrical cooperative (Potosí, Bolivia). Enabled on-site meter reading registration with GPS geolocation and automatic thermal receipt printing via Bluetooth. Backend built with **Laravel**, consuming the cooperative’s existing system REST APIs. Full management of Bluetooth peripheral lifecycle and runtime Android permissions.

- ▷ Field technician authentication and meter reading registration with GPS coordinates.
- ▷ Automatic pre-notice printing via **Bluetooth ESC/POS** thermal printer upon each reading.
- ▷ Full thermal printer management: device scanning, pairing, and automatic reconnection.
- ▷ Interactive client map rendered with **flutter_map + OpenStreetMap** (no API key required).
- ▷ REST API integration with the cooperative’s legacy system; local printer config persistence.

Process Automation with n8n

March 2026

Automation Projects (in collaboration)

- ▷ Automated bulk migration of **6,500+ Excel rows** into the ballot system with full traceability.

▷ Multi-source data integration and validation via **n8n** workflow orchestration.

Full Stack Developer — Church Information Website (IENAD BETANIA RESTAURACION)

2026 · In Production Use

Personal Project · Python · Streamlit

- ▷ Built and deployed a website for a local church, currently **in active use**, providing up-to-date information on church events.
- ▷ Includes links to the church's social media and relevant information about the congregation and its activities.
- ▷ Developed entirely with **Python + Streamlit** as a lightweight, easy-to-maintain solution.

FEATURED PROJECTS

Synapse — Comprehensive Educational Platform

2025 – 2026

Laravel 12 · PHP 8 · MySQL · JavaScript

- ▷ Full-featured LMS: course management, user administration and academic content delivery.
- ▷ Role-based access (Student / Teacher / Admin) with personalised dashboards per profile.
- ▷ Automated assessments, progress tracking, and detailed admin reports. Optimised with caching and eager loading.

Financial Fraud Detection with Deep Learning

2025 – 2026

Python · Streamlit · TensorFlow/Keras · Scikit-learn · Pandas · Plotly

- ▷ Hybrid **Isolation Forest + Autoencoder** system (20-12-8-12-20 architecture) for anomaly detection.
- ▷ Pipeline for imbalanced data (fraud <0.1%): **94.2% precision / 91.8% recall** on 150,000+ transactions.
- ▷ Interactive Streamlit dashboard with anomaly visualisation and adaptive risk scoring.

TECHNICAL SKILLS

Backend	Python, PHP, Laravel, Node.js, Express.js, Java, C++
Frontend	React, Astro, Flutter, HTML5, CSS3, JavaScript
Data / ML	Pandas, NumPy, TensorFlow, Keras, Scikit-learn, Streamlit
Databases	MySQL, PostgreSQL, MongoDB, SQLite, SQL Server
Tools	Git, GitHub, n8n, Docker, Linux, Jira

EDUCATION

Bachelor of Computer Engineering

2024 – Present

Universidad Mayor de San Simón (UMSS) — Cochabamba, Bolivia

CERTIFICATIONS & COURSES

- ▷ Intensive Python (NumPy, Pandas)
- ▷ Machine Learning (Microsoft)
- ▷ Recursion and Backtracking
- ▷ Competitive Programming — ICPC-UMSS
- ▷ Web Development with Laravel

ACHIEVEMENTS

- ✓ **Curricular Certificate:** Official Electoral Ballot System — Sub-national Elections Cochabamba 2026.
- ✓ **ICPC Cochabamba 2024:** Ranked **#17** in first competitive team participation.
- ✓ **Production Deployment:** ColcaTrufis live at moviruta.colcapirhua.gob.bo.
- ✓ **Successful Electoral System:** Actively used in the official Cochabamba 2026 electoral process.
- ✓ **Production Deployment:** Inventory system for Ferretería Urkupiña in active daily use.
- ✓ **Server Deployment:** Deployed and maintained multiple production systems on **Linux servers**.

LANGUAGES

Spanish	Native
English	B1